

ME2 — Voice Controlled Smart Device

Always-on keyword + intent recognition at the edge · no cloud round-trip

3 October 2026 · Crizepvill Dumalaog (Antigrav) · TinyDSCNN-48 · RTX 3050 (4GB) · Pi 5 Edge Validation

Model

Mic · 16 kHz · log-mel 40 × 251 (100 Hz frame rate, 2.5 s window)	
TinyDSCNN-48 · 4 DS-Blocks · 48-dim Depthwise-Separable CNN Two-Stage Wake Gating (0.95 cutoff) · Streaming Edge Inference	
Binary Wake Gate (0.95)	31-Class Intent & Slot Head
Actuator · GPIO RGB LEDs	Raspberry Pi 5 (ARM64)

Item	Value
Parameters / weights	0.028 M (27.6k) · 0.077 MB (76.7 KB INT8 ONNX)

Validation on the Raspberry Pi 5

Item	Value
Keyword / intent acc	100.0% wake / 71.6% intent (36.7% live)
False-accept rate	0.0% false wake (0/16) · 43.8% OOS (62.5% live)
Latency p95 / RTF	6.26 ms mean (7.23 ms p95) / 0.003 RTF
Runtime	onnxruntime 1.30.0 · 1 thread (ARM64)

Dataset

Item	Value
Source	ME2 Spoken Commands + Option B + Personal
Hours / utts	~7.4 h total / 10,682 train utts (+125 holdout)
Speakers	100+ reference · 7 personal · synthetic
Labels	19 intents · 31 leaf classes / slots

Training on NVIDIA RTX 3050 & INT8 PTQ

Item	Value
Cluster	1× NVIDIA RTX 3050 Laptop GPU (4 GB VRAM)
Objective	Cross-Entropy + Focal Loss · INT8 PTQ
Optimiser	AdamW (lr=1e-3, cosine decay, wd=1e-4)
Steps / loss	60 epochs (13,500 steps) / 0.082 loss (93.65% val)

To Be Submitted · Coursework Verification & Review Checklist

GitHub repository	cf dumalaog/Crizepvill-Dumalaog-AI-231 (public, MIT license)
Dataset location	Hugging Face: airimonda/vcm-benchmark (pinned commit ab39857)
Compute cluster	Local Workstation · 1× RTX 3050 (4 GB) · ~42 min wall-clock · seed=20261002
Model weights	deployment/current_vcm/models/ (intent_31_int8.onnx & wake_binary_int8.onnx, MIT)

What the reviewer will look for

#	Item	Status
1	Repo public, one-command reproduction	Verified (Start-VCM.bat)
2	Dataset licensed and citable (DOI)	Verified (MIT / HF open)
3	Training logs + final checkpoint committed	Verified (runs/ & ONNX)
4	Pi 5 latency reproduced by posted benchmark script	Verified (6.26 ms / 7.23 ms)
5	Held-out test set, unseen speakers	Verified (71.6% seed / 36.7% live)
6	Baseline of comparable size compared	Verified (vs Kiko on Pi)